Nuclear Astrophysics Lab v1.0.0
Calculation: Complete analysis
Reaction: 2H(p,gamma)3He
Temperature: 1.500000000e+7 K
Density: 1.500000000e+2 g/cm3
Model: Evaluated total rate with explicit abundance and energy accounting

Source: NACRE II 2H(p,gamma)3He; Xu et al. 2013; official table, accessed 2026-10-15; https://www.astro.ulb.ac.be/nacreii/data/2hpg3he.dat

INPUTS

Temperature: 1.500000000e+7 K
Density: 1.500000000e+2 g/cm3
Reduced mass: 6.666666667e-1 u
Selected dataset: nacre-dpg
Mass fraction 1: 7.000000000e-1 1
Mass fraction 2 (ignored for identical species): 1.000000000e-2 1
A1; A2: 1; 2
Deposited energy: 1.000000000e+0 MeV/event
Available energy: 1.500000000e+0 MeV/event
Gamow approximation: Explicitly enabled

RESULTS

Molar rate coefficient: 1.060000000e-2 cm3 mol^-1 s^-1
Pair-rate coefficient: 1.760171411e-26 cm3/s
Volumetric event rate: 5.026981989e+23 reactions cm^-3 s^-1
Characteristic depletion scale: 1.257861637e+2 s
Local specific heating: 5.369408721e+15 erg g^-1 s^-1
Approximate Gamow peak: 6.484440632e+0 keV
Published low evaluation bound: 8.800000000e-3 cm3 mol^-1 s^-1
Published high evaluation bound: 1.240000000e-2 cm3 mol^-1 s^-1

EQUATIONS

NA <sigma v> = selected evaluated rate; <sigma v> = rate / NA
n_i = rho X_i / (A_i u)
r = n1 n2 <sigma v> / (1+delta12)
tau = 1 / (n_partner <sigma v>)
epsilon = r Edep / rho (when Edep supplied)

NUMERICAL DIAGNOSTICS

All displayed values are finite or explicitly unavailable. Application v1.0.0.

UNCERTAINTY

Numerical error, nuclear-data uncertainty, temperature, density, and model approximation are separate. No combined uncertainty is available without covariance information. Unknown is not zero.
Published low/high evaluation bounds; no covariance. Not assigned a confidence level here.

ASSUMPTIONS

Fixed temperature and density, explicit mass fractions. No screening correction added.
Binary reactions only; no reverse rates, excitation-network correction, or evolution.

WARNINGS

Evaluated total rate, 60 tabulated temperatures
Deposited energy is an explicit input, not Q calculated from masses. Exclude escaping neutrinos and any nonlocal photon energy.
Single-channel local heating only; no reaction network, transport, hydrostatic equilibrium, or stellar evolution.
Slowly varying S(E), unscreened nonresonant charged particles. Not a universal effective energy for a real reaction.
Width is the FULL 1/e width of the quadratic Gaussian approximation, not FWHM or a hard integration cutoff.
Broad/asymmetric kernel: the Gaussian interval is only a rough guide.
Distinct species: r=n1 n2<sigma v>.
Homogeneous dilute plasma, binary reactions, fixed conditions; inverse reactions omitted.
For a+b with b a constant reservoir: dn_a/dt=-n_a n_b<sigma v>; n_a(t)=n_a0 exp(-t/tau).
This is a reaction depletion scale, not a radioactive decay half-life. Density and temperature are fixed.
Abundances use n_i=rho X_i/(A_i u): nuclear masses approximated by mass numbers. Remaining mass fraction is inert in this single-channel calculation.
No measured total sigma(E) grid is bundled for this selection. No thermal integration, S(E), or missing resonance result is inferred from a total rate.

This application provides educational and research-oriented nuclear astrophysics calculations under explicitly stated assumptions. It is not a complete stellar evolution or nucleosynthesis simulation.